

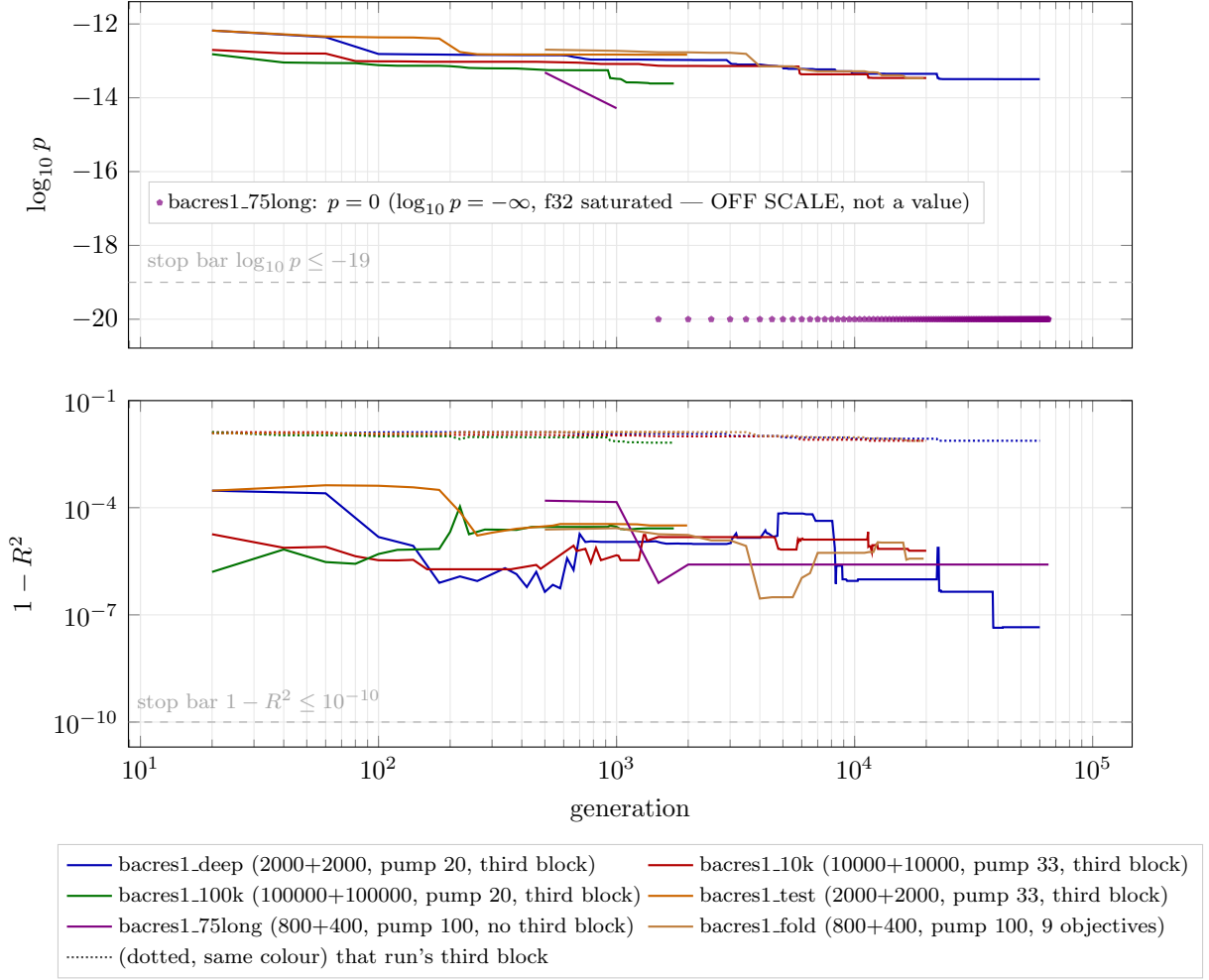


Figure 1: Convergence on the hyperspherical fitness function. $\log_{10} p$ (top) is $I_{\sin^2 \theta}((m - 1)/2, 1/2)$, in which the objective count m is consumed by the incomplete beta: p is a probability on $[0, 1]$, so $p = 10^{-19}$ means the same thing at 6 objectives, at 9 and at 19,000, and runs under different objective sets are directly comparable on this axis in a way no error metric allows. A FALLING p IS NOT BY ITSELF A RECOVERED LAW: the stop bar has two halves and both must pass, so $1 - R^2$ (bottom) is drawn beside it with the bar each run was actually run under. Markers on the floor of the top panel are p underflowed to exactly zero ($\log_{10} p = -\infty$) — the f32 angle saturated on the pole. They are off the scale, not at a value, and they are not success: read the panel below them. The dashed rules belong to the runs that recorded a bar; bacres1_10k, bacres1_test were written before the stream carried one, so no rule here is theirs. In the lower panel a SOLID line is train error and a DOTTED line of the same colour is that run's third block (SMOGD/SMOTE synthetic rows), which enters the HFF angle like any other objective. Validation error is measured and is in the data tables; it is left off the panel only because it tracks train closely on every run drawn here. Traces are thinned to their steps; no value is averaged and every point drawn is a beat the run held.